Deep Smoke Segementation

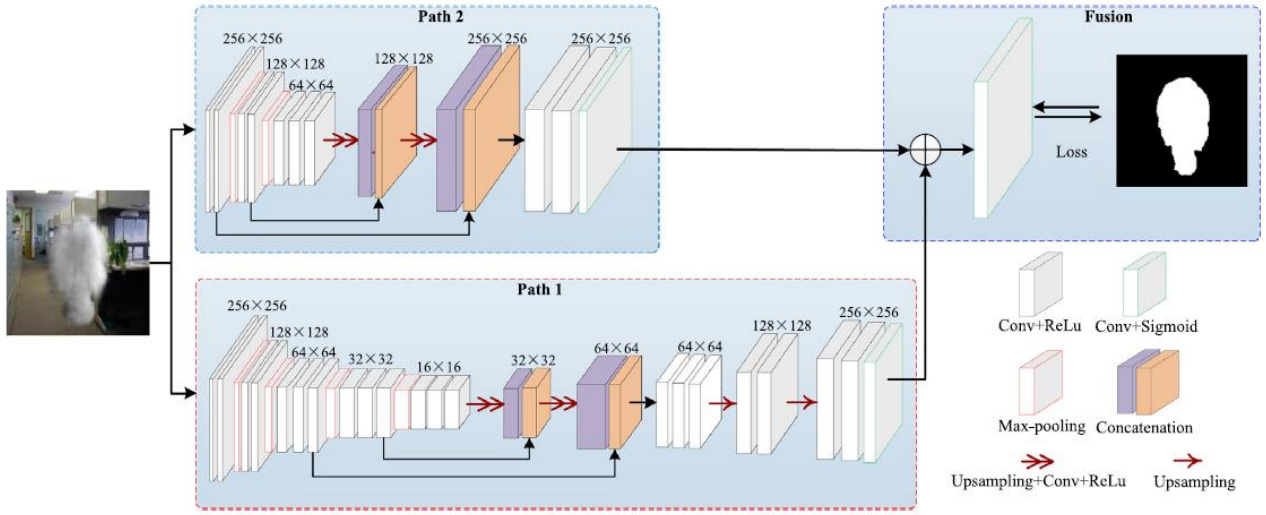
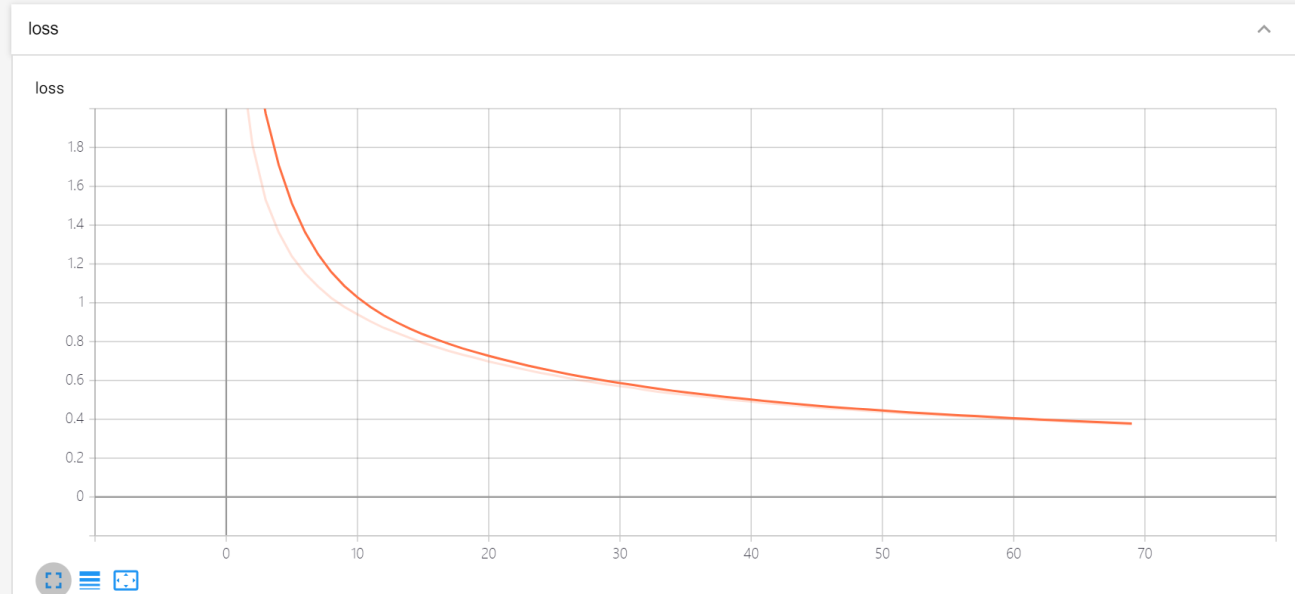


Fig. 3. Our network structure with two paths and skip layers.

利用两路unet的结构，深层获得全局语义信息，浅层获得细节信息

训练细节

- 2080ti训练了69轮。每轮2小时。训练数据集7w
- 最后输出经过sigmoid采用BCEloss
- optimizer = optim.SGD(model.parameters(), lr=1e-3, weight_decay=1e-5,momentum = 0.9)



袁非牛数据集SD01(RGBA)

$$\mathrm{Mse}_i = \frac{\sum_{k=1}^{h_i \times w_i} \left\| p_i^k - g_i^k \right\|^2}{h_i \times w_i}$$

In our experiments, we compute the a
dataset for performance evaluation:

$$\mathrm{mMse} = \frac{1}{N} \sum_{i=1}^{N} \mathrm{Mse}_i$$

	miou	mMse
袁(官方)	71.04	0.2745
我	65.7	0.2107

原图

Label

result

原因：
1.训练轮数不够多
2.图片没归一化
3.Label有问题
4.训练未数据增强

我的RGB（接下来的数据都是非官方）

	miou	mMse
RGB	0.796	0.1732

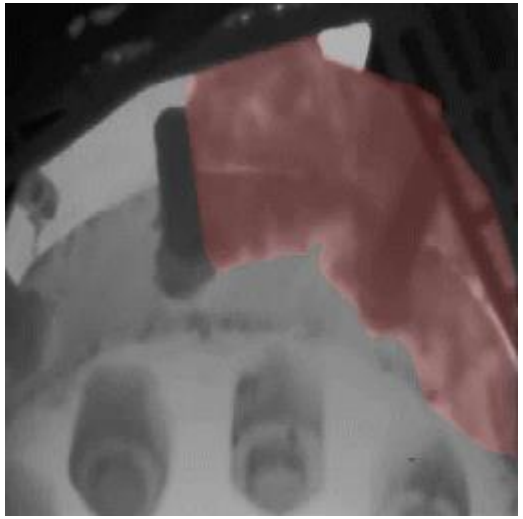
Label

result

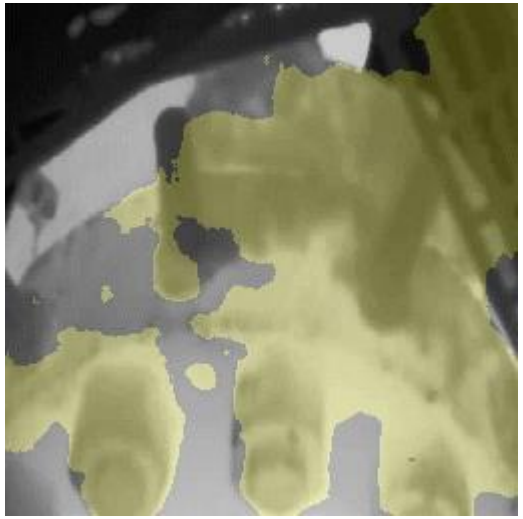
我的灰度图（转成单通道前的）

	miou	mMse
袁	0.796	0.1732
3DUnet	0.74	

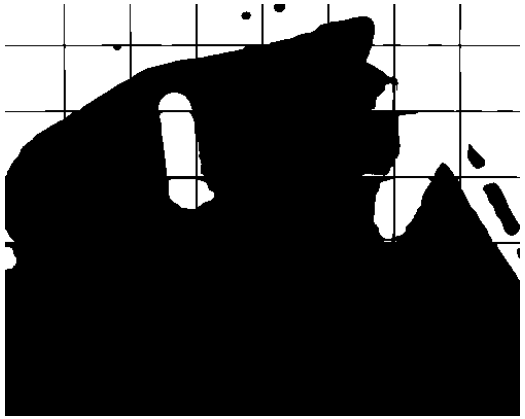
Label



result



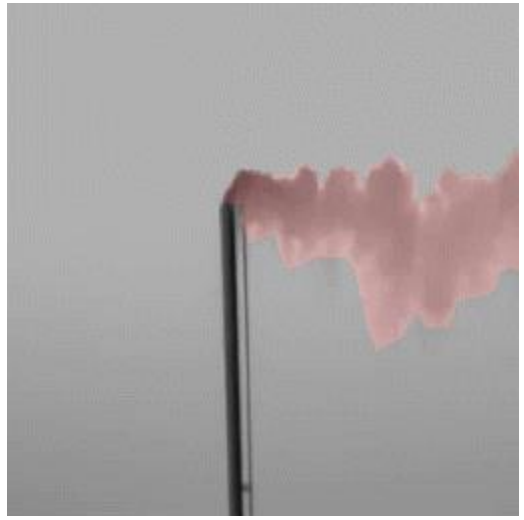
3D卷积



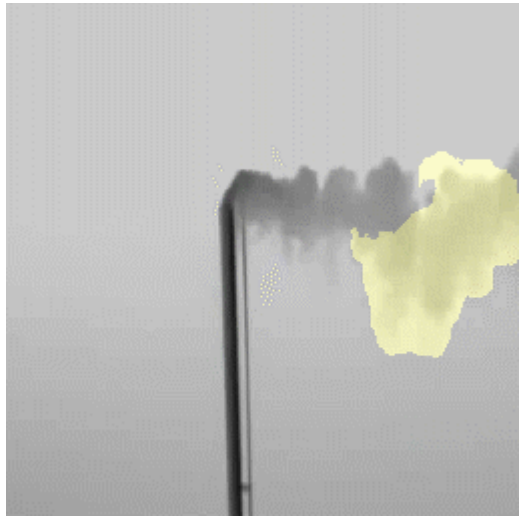
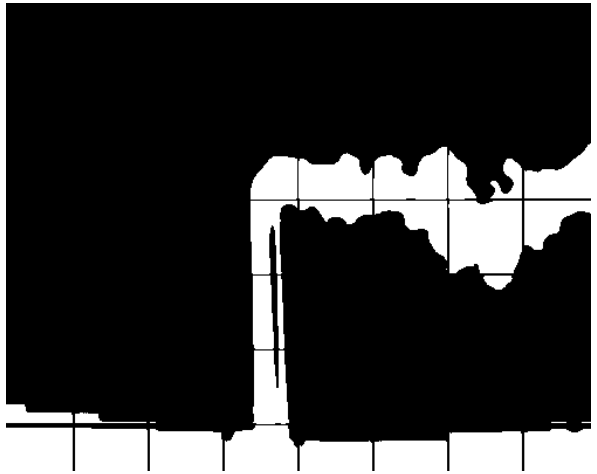
我的灰度图（转成单通道前的）

	miou	mMse
GRAY_idx17	0.8850	0.1087
3Dunet	0.69	

Label



result



3D卷积